



Healthcare diagnostics with an adaptive deep learning model integrated with the Internet of medical Things (IoMT) for predicting heart disease

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ARTICLE INFO

Keywords:

Heart Disease
Cardiovascular Diseases
Internet of Medical Things (IoMT)
Artificial Intelligence (AI)
Coronary Artery Disease
Forecast Model
TabNet
catBoost
Electrocardiograms (ECG)
Blood Pressure Readings
Multimodal Dataset

ABSTRACT

Heart disease is a widespread health problem that poses a significant global challenge. It includes a variety of heart-related conditions such as heart failure, coronary artery disease, and ventricular fibrillation all of which contribute to a significant global morbidity and mortality burden. Poor diet, physical inactivity, smoking, and underlying health conditions such as hypertension and diabetes. To mitigate the impact of this multifaceted problem, comprehensive preventive strategies, early detection through regular medical check-ups, and effective management are required. To promote heart-healthy habits and reduce the prevalence of this widespread health concern, not only individual awareness and behavioral changes are required, but also systemic efforts in healthcare policy, education, and accessible medical services. This study presents a novel coronary artery disease Forecast Model that seamlessly combines the Internet of Medical Things (IoMT) and Artificial Intelligence (AI) in response to the growing global burden of cardiovascular the approach uses advanced techniques, specifically TabNet in conjunction with catBoost, to improve the accuracy and efficiency of heart disease prediction. The IoMT is an important component because it enables seamless gathering of information from a wide range of healthcare devices, like wearables as well as sensors. diseases (CVDs). The approach uses advanced techniques, specifically TabNet in conjunction with catBoost, to improve the accuracy and efficiency of heart disease prediction. The IoMT is an important component because it enables seamless gathering of information from a wide range of healthcare devices, like wearable's as well as sensors. Proposed model employs Internet of Medical Things (IoMT) to collect a wide range of physiological data from connected medical devices, such as continuous heart rate monitoring, electrocardiograms (ECG), and blood pressure readings. TabNet and catBoost Artificial Intelligence (AI) algorithms are critical in processing and interpreting the complex data obtained through IoMT. TabNet, which is well-known for its ability to efficiently handle tabular data, is used for feature selection and extraction. CatBoost, a powerful gradient boosting algorithm, on the other hand, contributes to the model's predictive accuracy by handling categorical features and mitigating overfitting. The combination of these cutting-edge techniques improves precision as well as interpretability, allowing for a more detailed understanding of the factors influencing predictive outcomes. A large dataset with a variety of patient profiles, risk factors, and medical histories is used in the model's training regimen. By fine-tuning this diverse dataset, the model's adaptability to different demographic and clinical scenarios is improved, paving the way for personalized heart disease risk assessments. Furthermore, the proposed model supports immediate processing, allowing for rapid predictions based on streaming IoMT data. Finally, Heart Disease Prediction Model represents a

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<https://doi.org/10.1016/j.bspc.2024.105988>

Received 26 September 2023; Received in revised form 19 January 2024; Accepted 28 January 2024

Available online 17 February 2024

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paradigm shift in predictive healthcare by combining the strengths of IoMT and advanced computer vision techniques. The research envisions the implementation of this integrated model as a transformative tool in healthcare. By combining the real-time data acquisition capabilities of IoMT with the predictive prowess of TabNet and catBoost, the model aims to provide timely and accurate predictions of heart disease risks. This innovative approach holds great promise for ushering in a new era of personalized and effective preventive care strategies, ultimately leading to improved patient outcomes in cardiovascular health.

Nomenclature		GBDT	Gradient-Boosting Decision Tree
Acronym		EDCNN	Enhanced Deep Learning-Assisted Convolutional Neural Network
Description		CBO	Crow Browse Optimization
IoMT	Internet of Medical Things	CLSTM	Cascaded Long Short Term Memory
ECG	Electrocardiograms	iForest	Isolation Forest
AI	Artificial Intelligence	MDDS	Medical Decision Support Systems
CAD	Coronary Artery Disease	AIoMT	Artificial Intelligence of Medical Things
VF	ventricular rhythm disorders	EO	evolutionary Optimization
POC	Point-of-Care	GLU	Gated Linear Unit
HLDA-MALO	Hybrid Linear Discriminant Approach With Altered Ant Lion Optimization	SVM	Support Vector Machine
IoT	Internet of Things	HSC	Hierarchical Self-Calibrating
ML	machine learning	RCNN	Region Convolutional Neural Network
MQTT	Message Queuing Telemetry Transport		

1. Introduction

Heart disease, which includes conditions such as coronary artery disease (CAD), heart failure, as well as ventricular rhythm disorders (VF), is a significant global health issue.. This diverse range of cardiovascular diseases helps significantly to the global morbidity and mortality burden. Poor dietary habits, sedentary lifestyles, smoking, and underlying health conditions such as hypertension and diabetes all play important roles in the development of heart disease. Preventive measures such as leading a healthy lifestyle, managing risk factors, and seeking medical advice for early detection and treatment are critical in lowering the burden of heart disease. Public health campaigns and awareness programs are critical for informing people about the importance of heart health and encouraging healthy lifestyle choices. Medical research and treatment modalities advancements also contribute to better outcomes for people with heart disease. [1]. The investigation of IoMT's role in healthcare highlights its promising impact on advancing medical services and optimizing healthcare systems for better outcomes [2].

To address this multifaceted health issue, a multifaceted approach is required, including comprehensive preventative measures, early detection through regular medical check-ups, and effective disease management. Promoting heart-healthy habits necessitates not only individual awareness and behavioral changes, but also systemic efforts in healthcare policies, educational initiatives, and the availability of medical services. This method allows healthcare professionals to detect potential problems early on and intervene before they become more serious [3]. This decision support system can empower local healthcare providers and community health workers to monitor vital signs, recognize early symptoms, and initiate preventative measures by leveraging technology such as wearable devices and machine learning algorithms. This approach not only bridges the access gap to cardiac expertise, but it also has the potential to save lives by enabling proactive management of heart conditions in resource-limited settings, where timely diagnosis is critical to avoiding fatal outcomes [4]. In response to the growing global burden of cardiovascular diseases (CVDs), this study presents a groundbreaking Coronary Artery Disease Forecast Model.

The healthcare paradigm is undergoing a revolutionary

transformation as a result of the IoMT, and its application to tracking heart data represents a groundbreaking frontier. This novel IoMT model for heart data monitoring integrates a variety of medical devices, wearables, and sensors that convey and transmit real-time cardiovascular data [5]. The interconnected ecosystem collects vital information such as heart rate, cardiac recordings (ECG), blood pressure, and other pertinent metrics to provide a complete picture of a person's cardiac health. This IoMT model, which employs advanced data analytics and predictive methods, aids in the early detection of deviations and trends related to cardiovascular diseases. By utilizing them, healthcare providers may acquire an improved awareness of their patients' conditions. [6].

This convergence of cutting-edge techniques not only improves precision but also interpretability, allowing for a more nuanced understanding of the factors influencing predictive outcomes. Training the model on a diverse dataset that includes a wide range of patient profiles, risk factors, and medical histories ensures its adaptability to various demographic and clinical scenarios, allowing for personalized heart disease risk assessments. Furthermore, the model supports instant processing, allowing for rapid predictions based on streaming IoMT data. This groundbreaking Heart Disease Prediction Model represents a paradigm shift in predictive healthcare, capitalizing on the strengths of IoMT and advanced AI techniques. The IoMT integration enables remote chronic disease monitoring and diagnosis, transforming healthcare delivery. IoMT allows for continuous real-time patient monitoring, enabling timely interventions and personalized treatments. IoMT improves patient care while also lowering healthcare costs by fostering remote healthcare capabilities [7]. It is envisioned as a game-changing tool in healthcare, with the goal of providing timely and accurate predictions of heart disease risks, ushering in an era of personalized and effective preventive care strategies. This novel approach has a lot of potential.

The paper is structured as follows: Section 2 deliberates the review on the related works on heart disease prediction models. The working procedure of the proposed model is explained in Section 3. The result analysis and the comparisons are given and explained in Section 4. Section 5 concludes the paper with some directions for future research.

2. Related Works

Manickam P et al (2022) [8]. This comprehensive review examines the integration of machine learning with the World Wide Web of Medical Things, highlighting its critical role in enhancing the capabilities that underlie contemporary medical sciences. The synergistic relationship between AI and IoMT is discussed, with particular emphasis on advances in cardiac measurement, cancer diagnosis, and diabetes management. The review emphasizes how AI, including machine learning and deep learning, contributes to IoMT device efficiency by simplifying clinical diagnosis and decision-making processes. The discussion expands to AI's use in assisting advanced robotic surgeries, demonstrating its transformative impact on advanced biomedical procedures. The review critically examines the role of AI in improving IoMT device functionality, detection accuracy, and risk assessment, while also addressing technological challenges. Finally, the potential for tailored IoMT devices based on AI is investigated, providing insights into the development of effective Point-of-Care (POC) medical equipment for the intelligent medical landscape.

Shah D et al (2020) [11]. This research paper focuses on the pressing problem of heart disease, emphasizing the critical need for accurate and timely detection in light of its global prevalence and impact on mortality. The study investigates the use of algorithms for supervised learning such as Bayes's choice tree, k-nearest neighbor, and random forest on a dataset from the UCI repository's city of Cleveland database, which contains 303 instances and 76 attributes, using data mining techniques. The study focuses on 14 important traits to predict patients' risk of developing heart disease. According to the findings, the K-nearest neighbor algorithm has the greatest accuracy score among those tested, emphasizing its promise as a successful instrument for forecasting and controlling coronary artery disease.

Manimurugan S et al (2022) [9]. The Internet of Things, or IoT for short, specifically the IoMT, is used in this study to diagnose and outlook heart disease. In the first stage, the suggested approach classifies data from sensors in hospitals using a hybrid linear discriminant approach with altered ant lion optimization (HLDA-MALO). If the first stage correctly foresees coronary artery disease, the subsequent phase, which focuses on echocardiogram image classification using hybrid model, may be unnecessary. The findings from both stages are combined in the study to consolidate and validate heart disease predictions. This innovative approach shows the power of IoT and advanced analytics classification techniques in improving medical diagnostics and prognostics for heart conditions.

Oniani S et al (2021) [10]. This paper examines the Internet of Things (IoT), Big Data, and machine learning (ML) in the context of customized healthcare in depth. The focus is on AI methods used in various health domains, including heart disease diagnosis, modeling for prediction, robotic surgery, and personalized treatment. The research emphasizes the success of key AI methods in evaluating patient data to improve health conditions, such as the k-nearest-neighbors method, support vector machines, regression modeling, and tree classification. Notably, robotic surgery systems like Transoral Surgical Robotics and the Automated Endoscopic System for Optimal Positioning provide significant advantages such as fewer invasive procedures with fewer bleeds and faster recovery. Furthermore, the Internet of Medical Equipment emerges as an important component, addressing a wide range of health issues via ongoing tracking of biological settings, managing diabetes, and medical decision support.

Patel W et al (2021) [11]. This novel approach addresses access to healthcare services by leveraging the transformative power protocol. Proposed system incorporates IoT devices to collect critical physiological data from patients and transmit it to the cloud for analysis. ML improves decision-making and prediction, particularly in heart disease prognosis. Message Queuing Telemetry Transport (MQTT) allows for seamless communication between devices, resulting in an interconnected healthcare ecosystem. The solution provides real-time heart

condition predictions, particularly when combined with patients' medical profiles, allowing for informed decision-making and timely emergency alerts. The system's comprehensive evaluation, which took into account metrics such as accuracy, sensitivity, and specificity, indicates that it has promising potential for improving cardiovascular disease diagnosis and intervention.

Yuan X et al (2021) [12]. This article introduces a machine learning-based model based on the Fuzzy-GBDT algorithm to address the critical need for accurate heart disease prediction and severity determination. By integrating fuzzy logic and a gradient-boosting decision tree (GBDT), the algorithm improves binary categorization prediction while guaranteeing robustness and reducing data complexities. Furthermore, to avoid overfitting, a combination of Bagging with Fuzzy-GBDT is proposed. In binary and multiple classification predictions, the resulting Bagging-Fuzzy-GBDT model outperforms other models, demonstrating remarkable accuracy and stability. This novel approach has the potential to improve the accuracy of heart disease monitoring, prediction, and diagnosis, leading to earlier intervention and better healthcare outcomes.

Pan Y et al (2020) [13]. The Enhanced Deep Learning-Assisted Convolutional Neural Network (EDCNN) is proposed in this study to address the complicated nature of coronary artery disease diagnosis by leveraging advances in deep learning as well as the IoT. By combining multi-layer perceptron and regularization learning approaches, the EDCNN model aims to improve patient prognostics. The impact of full and minimized features on classifier productivity in terms of execution time and accuracy is demonstrated. The EDCNN system, which is based on the IoMT, is a decision support tool that enables accurate diagnosis of heart disease all over the world. A comparison with traditional methods displays the diagnostic system's effectiveness in determining heart disease risk levels.

Adewole et al (2021) [5]. The present section investigates the revolutionary potential of Web of Things technology to enhance personalized healthcare services, particularly in the context of the IoMT. While IoT integration in healthcare generates massive amounts of data, cloud-based applications play a critical role in addressing secure storage and efficient service delivery. Despite significant investments in healthcare, the industry faces challenges because many institutions still use paper-based medical records, which impedes information sharing and decision-making. This impedes patient safety progress and contributes to the high prevalence of chronic diseases, including CVD. The chapter emphasizes the importance of effective prediction, diagnosis, and management strategies in light of rising CVD rates worldwide. As key components, the fundamental concepts of cloud computing, IoT, and IoMT are discussed.

Mansour et al (2021) [14]. This study presents an innovative diagnosis model for heart disease and diabetes in the context of smart healthcare, leveraging the convergence of AI and the Internet of Things (IoT). The proposed models combine the Crow Browse Optimization (CBO) algorithm with Cascaded Long Short Term Memory (CLSTM) for improved diagnostic accuracy. Wearables and sensors enable seamless data acquisition, while CSO optimization improves the CLSTM model's classification capabilities. The Isolation Forest (iForest) is a technique for removing outliers and refining diagnostic results. Experimental validation reveals an accuracy of 96.16 % for heart disease and 97.26 % for diabetes. This convergence-based CSO-LSTM model appears to be a promising tool for disease diagnosis in smart healthcare systems.

Merabat A et al (2023) [15]. In the rapidly evolving healthcare landscape, this paper highlights the rapid development of ML and the IoMT in the realm of Medical Decision Support Systems (MDSS). 73 % of healthcare providers believe AI has the potential to improve individual performance, according to a survey conducted by Intel and Converged Technologies. By leveraging real-time patient data from IoMT devices, AI becomes a critical tool in assisting medical decision-makers. This overview examines how AI and IoMT are being integrated into healthcare systems, with a focus on their roles in prediction, diagnosis, and

treatment recommendation. The paper introduces novel methodologies for improving the accuracy, efficiency, and effectiveness of medical decision-making, indicating progress and laying the groundwork for future research and innovation in the field.

Rajeswari S et al (2022) [16]. This chapter examines the revolutionary effects of the World Wide Web (WWW) of Things and AI on the medical field, providing an in-depth look at their integration. It is being investigated how IoT and AI can be used in medical devices such as smartwatches, connected gadgets, and smart security systems. Data security, power requirements, and scalability are among the issues addressed, paving the way for potential research solutions. The chapter delves into real-world applications and case studies to demonstrate the advancement of IoT and AI in biomedical settings. As a recent global challenge, the COVID-19 pandemic is discussed, emphasizing the role of AI in diagnosing and monitoring health crises. The chapter concludes by emphasizing the potential of AI and IoT to reshape healthcare by providing solutions for remote monitoring, disease detection, and treatment.

Fahim et al (2024) [17]. This section looks at how AI is being integrated into medical devices, giving rise to the concept of Artificial Intelligence of Medical Things (AIoMT). The global acceptance of AI applications in healthcare increased in response to the COVID-19 pandemic, demonstrating the potential of intelligent technologies to save lives. AIoMT enables the collection, analysis, and interpretation of massive amounts of patient data through the use of smart devices and technologies such as big information and the cloud. The chapter provides a thorough examination of electronic devices in AIoMT, focusing on their roles in transforming traditional healthcare into a personalized and efficient experience. It addresses challenges such as data security and interoperability, as well as open research issues for AIoMT's future advancement, ensuring improved medical standards and patient care.

Poomari Durga K et al (2022) [18]. This study delves into the complexities of health issues, with a particular emphasis on heart disease monitoring techniques to address complications and improve overall well-being. To achieve high accuracy, the proposed methodology employs various machine learning techniques on the benchmark University of California Irvine Dataset. The main goal is to improve human health and reduce mortality rates. The integration of the World Wide Web of Medical Things, or IoMT, into a Cardiac Monitor System, enables remote health monitoring and alerts healthcare professionals to abnormalities. This system collects and stores patient data via sensors, allowing medical specialists cloud-based access. While prevalent in developed countries, the proposed remote cardiac monitoring system highlights the need for implementation in underdeveloped regions.

Malibari A et al (2023) [19]. This paper uses IoT technology to tackle the challenging challenge of anticipating chronic diseases such as coronary artery disease and kidney disease. The suggested EO-LWAMCNet model, designed with evolutionary Optimization (EO), leverages data from sensors collected from patients and transmitted to the cloud to make accurate predictions. Using CKD and HD datasets, the model is trained and tested, and it accomplishes high accuracy rates of 93.5 % and 94 %, respectively. Metrics such as accuracy, MCC, F1-score, and miss rate are used to assess the model's ability to predict either the presence or absence of chronic diseases while reducing classification errors.

Awotunde et al (2023) [20]. This chapter investigates the transformative potential of AI in the Internet of Medical Things (IoMT) to improve disease monitoring, diagnosis, and treatment. In response to data management challenges in the IoMT, the integration of AI-based self-adaptive strategies is proposed, potentially revolutionizing the healthcare sector. The IoMT, which is equipped with embedded sensors, generates massive datasets, necessitating efficient data exchange, analysis, and patient monitoring strategies. The role of AI in collecting, scrutinizing, and interpreting big data stored in cloud databases allows for more precise disease prediction while reducing the burden on healthcare systems. The collaboration of AI and IoMT not only speeds up

diagnostics but also opens up opportunities for better disease management and vaccine production. Despite its transformative potential, the chapter acknowledges research challenges and emphasizes the importance of ongoing growth and development. The proposed system offers a revolutionary approach to heart disease prediction by integrating TabNet and CatBoost, leveraging their strengths in structured data analysis. This model ensures interpretability, allowing healthcare professionals to comprehend influential factors. Its IoMT foundation enables real-time monitoring, fostering personalized healthcare interventions and proactive management for improved cardiovascular outcomes.

3. Proposed system

To address the global challenge of heart disease, the proposed system is a novel coronary artery disease Forecast Model that integrates the technologies. The proposed study will analyze clinical datasets using machine learning techniques to address the critical challenge of heart disease prediction. The goal is to identify the key characteristics and factors that contribute to accurate predictions in the field of heart disease using advanced algorithms. Improve the precision and effectiveness of predictive models by identifying these critical features, ultimately contributing to improved diagnostic capabilities in the context of heart disease [21]. The model collects a wide range of physiological data from connected medical devices, including continuous heart rate monitoring, electrocardiograms (ECG), and blood pressure readings, by leveraging the capabilities of IoMT. Advanced AI algorithms, specifically TabNet and catBoost, are used to improve data processing efficiency, feature selection, and predictive accuracy. The model makes use of a large dataset that includes various patient profiles and risk factors, allowing it to adapt to different demographic and clinical scenarios. The proposed system, with its real-time processing capabilities, aims to provide timely and accurate predictions of heart disease risks. This integrated approach is transformative. This integrated approach is a game changer in healthcare, promoting personalized and effective preventive care strategies to improve cardiovascular health outcomes.

Fig. 1 depicts the comprehensive architecture created for Heart Disease Prediction, which makes use of an advanced integration method. The diagram depicts the key components and processes of this ground-breaking healthcare forecasting model.

3.1. Data preprocessing for heart disease Prediction

Robust data preprocessing is critical in the field of Heart Disease Prediction to ensure the quality and relevance of the collected data. The process includes meticulous cleaning, filtering, and feature extraction techniques that improve the predictive model's overall efficiency and accuracy.

3.1.1. Cleaning collected Data

(a) Missing data Handling

Imputation Techniques: Addressing missing values is imperative to maintain the integrity of the dataset. Various imputation techniques, such as mean or median imputation, forward or backward filling, or employing more advanced methods like k-nearest neighbors imputation, are applied based on the nature and extent of missing data.

Exclusion of Incomplete Entries: In scenarios where missing data is extensive or the pattern is irregular, exclusion of incomplete entries may be considered. This ensures that the dataset used for analysis is consistent and complete.

(b) Outlier detection and removal

Identification of Outliers: To identify outliers in a dataset, robust statistical approaches or artificial intelligence algorithms are used. Outliers are points of data that stray substantially from the overall pattern, potentially biasing the analysis.

Visualization Techniques: Tools like box plots scatter plots, or Z-score analysis aid in visually identifying outliers, allowing for a

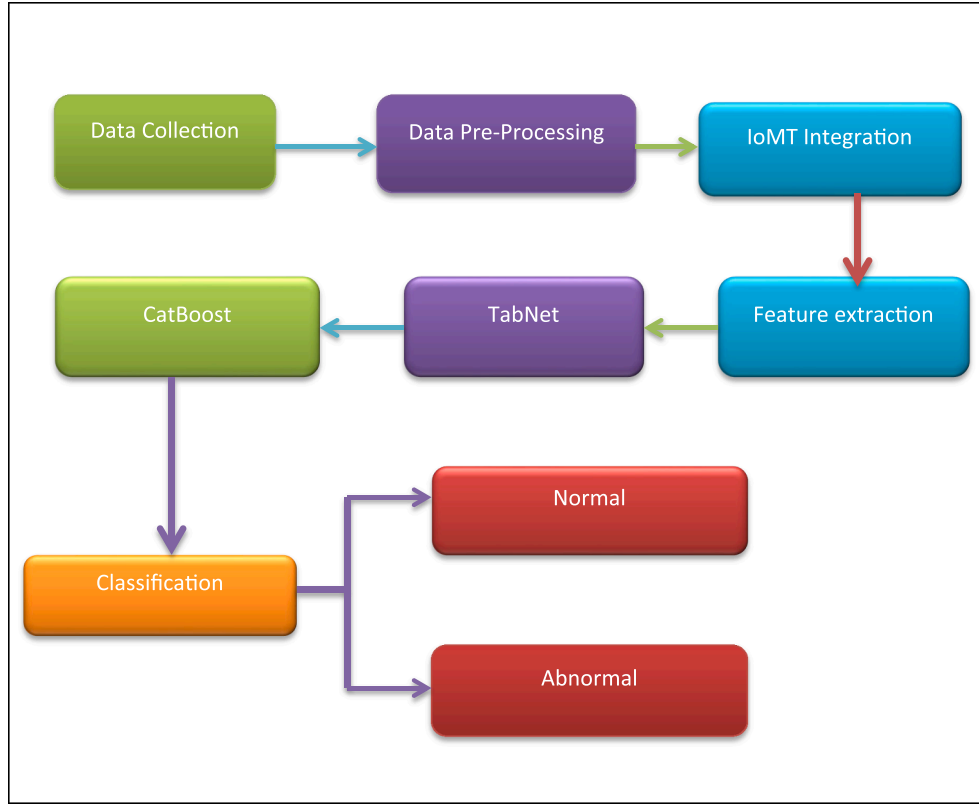


Fig. 1. Block Diagram for Heart Disease Prediction.

comprehensive understanding of their impact.

Anomaly Removal Strategies: Once outliers are identified, removal strategies are implemented to eliminate their influence. This may involve outright exclusion, transformation, or replacing outlier values with more suitable alternatives.

3.1.2. Filtering

Filtering is an important step in the data preprocessing phase for heart disease prediction because it improves the consistency and reliability of the collected dataset. This entails taking strategic steps to reduce noise and ensure lines across chronological and geographic dimensions.

(a) Noise Reduction

Physiological parameters frequently have inherent noise, which can impair the interpretation of data and forecasting accuracy. Filters are a fundamental technique for reducing noise and ensuring a more coherent dataset. Here are some of the most important aspects of noise reduction in physiological data. Median filters are especially useful for removing impulse noise because they replace each data point with the median value in its neighborhood. This aids in reducing noise-induced spikes.

$$y[n] = \text{median}(x[n - \lfloor W/2 \rfloor], \dots, x[n], \dots, x[n + \lfloor W/2 \rfloor]) \quad (1)$$

(b) Signal processing Techniques

Using methods for signal processing such as Fourier analysis or wavelet transforms assists in recognizing and reducing noise components, thereby improving the clarity of essential features. Fourier analysis it aids in comprehending the frequency-related components of a signal. A signal's Fourier transform $X(f)$ is given by:

$$X(f) = \int_{-\infty}^{\infty} x(t) \cdot e^{-j2\pi ft} dt \quad (2)$$

(c) Normalization and scaling in data preprocessing

Normalization and scaling are critical steps in preprocessing data because they ensure that the data is in a format suitable for learning

algorithms. Here are some important aspects of normalizing and scaling: The goal is to have numerical values for different features on a similar scale. Scaling keeps particular variables from dominating the process of learning solely because of their greater magnitude. Scaling methods that are commonly used include Min-Max scaling, the Z-score normalization, and robust scaling, each with its own set of advantages depending on the data distribution.

(d) Categorical Encoding

For features with categorical variables, encoding is performed to convert them into numerical equivalents. Many machine learning algorithms require numerical input, and encoding facilitates the incorporation of categorical data into the training process. Techniques like one-hot encoding or label encoding are employed, depending on the nature of the categorical variables.

Min-Max Scaling

$$X_{\text{normalization}} = \frac{X - \min(X)}{\max(X) - \min(X)} \quad (3)$$

Z-score Normalization

$$X_{\text{normalized}} = \frac{X - \mu}{\sigma} \quad (4)$$

Robust Scaling

$$X_{\text{normalized}} = \frac{X - \text{median}(X)}{IQR(X)} \quad (5)$$

(e) One-Hot Encoding

Converts categorical variables to binary vectors, generating a binary column for each. If a variable has three categories (A, B, C), it generates three binary columns, one for each category, each representing the presence or absence of the corresponding category.

3.2. IoMT Integration

The framework for heart data collection is an innovative strategy that combines the power of the IoMT with advanced technologies to comprehensively track and analyze cardiovascular health. The study intends to thoroughly examine the existing body of research in order to determine the effectiveness of IoMT in providing benefits to both patients and the healthcare system. Furthermore, the research will provide a concise overview of complementary technologies that supplement IoMT and shed light on the challenges encountered in the development of a smart healthcare system [22]. This framework captures real-time data such as heart rate, cardiac electrocardiograms (ECG), and blood pressure through the use of interrelated medical devices, wearables, and sensors, resulting in a dynamic and continuous monitoring system [23]. The IoMT model excels at the early identification of anomalies and trends related to heart diseases by leveraging artificial intelligence and machine learning algorithms. The use of a diverse dataset that includes various patient profiles improves adaptability across demographic information and clinical scenarios. This model’s immediate processing capabilities enable rapid forecasts based on streaming IoMT data, paving

the way for prompt actions and improved patient outcomes.

The IoMT model depicted in Fig. 2 is specifically designed for heart data monitoring. This comprehensive model combines a variety of technological components to create a streamlined and efficient system for monitoring and managing heart-related data. The IoMT model is built around interconnected devices, sensors, and communication protocols that allow for the continuous collection of vital cardiac data. By continuously collecting real-time medical data via wearable sensors machine learning models revolutionizes healthcare. By deploying wearable devices equipped with advanced sensors on key body parts like the chest, right wrist, and left ankle, the model captures motion parameters, ECG measurements, and other relevant metrics. The integration of remote monitoring technologies involves strategically placing sensors on specific body parts, such as the chest, right wrist, and left ankle, to measure motion parameters and gather real-time medical data. This interconnected sensor network forms the backbone of the IoMT model, enabling continuous monitoring and analysis of cardiovascular health for improved patient outcomes. This collaboration makes it easier to create predictive models for early disease detection and personalized healthcare interventions. The combination of IoMT and machine

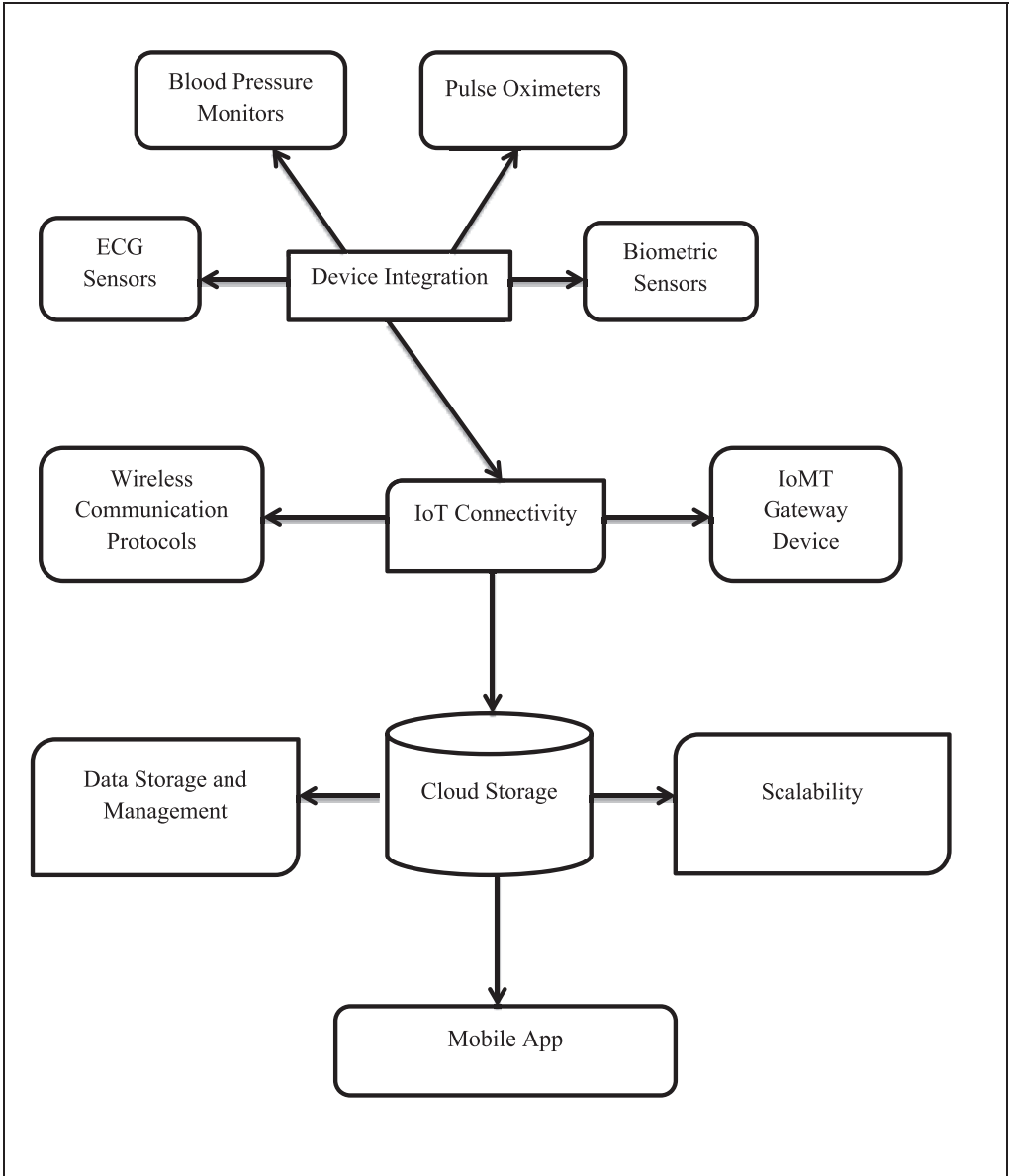


Fig. 2. IoMT Model for Heart Data Monitoring.

learning is a game changer, enabling proactive and individualized health management for better patient outcomes [24]. The model starts with the deployment of wearable devices outfitted with advanced sensors capable of measuring key physiological parameters like heart rate, ECG signals, and other relevant metrics.

3.2.1. Device Integration

The device integration and IoT connectivity components within the IoMT framework play a pivotal role in revolutionizing heart disease monitoring, offering a comprehensive and personalized approach. In this section, we will discuss these aspects more efficiently, highlighting the significance of each device and the role of IoT connectivity in ensuring seamless data transmission and analysis:

ECG Sensors: Wearable devices equipped with ECG sensors are at the forefront of continuous heart monitoring. These sensors capture electrical signals of the heart, providing a detailed and real-time understanding of cardiac rhythms. This real-time data is essential for early anomaly detection and personalized interventions.

Blood Pressure Monitors: The integration of IoT-enabled blood pressure monitors adds a critical dimension to cardiovascular health assessment. These monitors not only measure blood pressure in real-time but also transmit this data seamlessly. The continuous monitoring of blood pressure contributes valuable information, aiding in the assessment of overall cardiovascular health.

Pulse Oximeters: Inclusion of pulse oximeters in the IoMT setup enhances the understanding of cardiac function. These devices monitor oxygen saturation levels in the blood along with heart rate, providing a more comprehensive set of metrics. This contributes to a nuanced analysis of cardiovascular health and aids in the early identification of potential issues.

An innovative exploration involves the integration of smart textiles embedded with biometric sensors in smart clothing. These sensors operate invisibly, collecting physiological data without intrusion. This non-intrusive and continuous monitoring of various heart-related metrics adds a layer of personalization to the monitoring system. The convergence of these devices creates a connected ecosystem, fostering proactive healthcare management and early detection of potential heart-related issues.

3.2.2. IoT Connectivity

Wireless communication Protocols

The utilization of secure and standard wireless communication protocols, such as Bluetooth Low Energy (BLE) or Wi-Fi, is fundamental. These protocols facilitate seamless connectivity, allowing devices to transmit data to the central hub in real-time. The choice of these protocols ensures not only efficiency in data transmission but also adherence to security standards, a crucial aspect in healthcare data management.

IoMT gateway Device

The introduction of the IoMT gateway device enhances the overall system architecture. This central hub collects data from various sensors employed in heart monitoring. Its role is crucial in securely and reliably transmitting this data to a cloud-based platform for comprehensive analysis, interpretation, and storage. The IoMT gateway acts as a bridge, ensuring that the data collected from diverse devices is effectively centralized and made available for further insights. In summary, the device integration and IoT connectivity components within the IoMT framework collectively form a sophisticated and interconnected system for heart disease monitoring. The seamless collaboration of devices, coupled with robust connectivity protocols and the IoMT gateway device, creates a dynamic ecosystem that holds immense potential for proactive healthcare management and early detection of cardiovascular issues.

3.2.3. Cloud-Based Platform

Create a strong cloud-based platform designed for the safe storage

and management of collected cardiovascular data. For authorized users, this platform ensures data integrity, confidentiality, and accessibility. The platform provides scalable storage solutions by leveraging cloud infrastructure, allowing for seamless archival, retrieval, and management of various heart-related metrics. Meeting these requirements is critical for handling the vast and complex datasets inherent in the healthcare domain [25]. Implement real-time processing capabilities in the cloud-based platform to allow for rapid analysis and response to critical events. This real-time processing feature ensures the timely identification of anomalies or emerging health issues, contributing to proactive healthcare interventions and improving the IoMT system's overall responsiveness. Create the cloud-based platform with scalability in mind, so that it can easily expand and accommodate an increasing number of connected devices and users. Scalability is critical for dealing with the growing volume of cardiovascular data generated by an expanding network of sensors and wearables. This scalable architecture ensures that the IoMT system maintains its performance, responsiveness, and adaptability as it evolves.

3.2.4. Apps for mobile Health

Connect Bluetooth-enabled health devices by integrating smartphone applications with IoT capabilities. These health apps are critical in gathering, processing, and transmitting heart-related data from various IoT-enabled devices. These mobile health apps allow users to conveniently monitor their cardiovascular metrics, providing a seamless and accessible way of tracking heart health. These apps facilitate real-time data acquisition by connecting to devices such as blood pressure cuffs or glucometers, providing users with a comprehensive view of their cardiovascular well-being. IoT technology integration improves the functionality of mobile health apps, resulting in a more holistic and user-friendly approach to heart health monitoring.

3.3. Feature Extraction

Feature extraction for heart disease involves identifying and selecting relevant attributes or characteristics from the raw data collected through various sensors or diagnostic tests. Feature selection, an essential component of automated data preprocessing in machine learning, entails the careful curation of an object's or dataset's most important and relevant features. This process prevents the potential degradation of classification system performance by systematically discarding redundant or irrelevant features. The use of feature selection not only improves accuracy by focusing on key variables, but it also helps to reduce the overall computational cost of modeling [26]. The goal is to capture essential information that can contribute to accurate prediction, diagnosis, or monitoring of heart-related conditions. Here are some common features extracted for heart disease analysis:

3.3.1. ECG Features

R Peak Detection is a critical step in the processing of ECG signals in which the algorithm identifies the prominent R-peaks that correspond to ventricular depolarization. These peaks represent the maximum electrical activity during each heartbeat. These real-time HR monitoring aids in the assessment of cardiac health and the detection of irregularities. ECG signals are used by doctors to diagnose conditions like arrhythmias, ischemic heart disease, and myocardial infarctions. The distinct waveforms and patterns in ECG readings provide important information about the health and functionality of the heart's electrical system. Because of the ease with which ECG signals can be obtained, it has become a standard diagnostic tool in both routine clinical assessments and emergency situations [27]. Furthermore, ST-Segment Analysis examines deviations in the ECG waveform's ST-segment. Changes in this segment can indicate conditions like ischemia or myocardial infarction, allowing for early detection and treatment of potential cardiovascular issues. These analyses contribute to a more comprehensive understanding of the cardiac cycle, allowing for more timely clinical

assessments and preventive measures.

3.3.2. Blood pressure Features

The gathering of vital parameters such as systolic and diastolic pressure is part of blood pressure characteristics. The higher of the two values, systolic pressure, reflects the force exerted on arterial walls during heart contraction. The lower value, diastolic pressure, represents the pressure between heartbeats when the heart is at rest. Continuous monitoring allows for a dynamic assessment of these pressures, providing insights into how the cardiovascular system works. Analyzing systolic and diastolic blood pressure values aids in the detection of hypertension, hypotension, or irregularities, thereby facilitating proactive healthcare interventions and personalized treatment strategies.

3.3.3 Clinical parameters

LDL, or low-density lipoprotein, and higher-density lipoprotein (HDL) cholesterol levels are important markers for heart conditions. Elevated LDL cholesterol, also known as “bad” cholesterol, is dangerous, whereas higher levels of HDL cholesterol, also known as “good” cholesterol, are generally beneficial. Monitoring cholesterol levels can help identify potential cardiovascular risks and guide lifestyle or medical interventions.

Similarly, monitoring blood sugar levels is critical, especially in diabetics. Glucose fluctuations can have an impact on heart health and contribute to cardiovascular complications. Continuous blood sugar monitoring enables proactive management, lowering the risk of diabetes-related heart problems. These parameters provide valuable insights for customized medical care and preventive measures when taken together.

3.3.4. Demographic and medical History

Age, gender, and family history are all important factors in determining the disease. Age is important because the likelihood of developing cardiovascular issues increases with age. Furthermore, gender differences in the prevalence and manifestation of heart-related conditions may exist.

Family history can reveal genetic predispositions and hereditary factors that can contribute to heart disease. Incorporating these factors into the analysis improves the predictive accuracy of heart disease models, allowing for tailored interventions and proactive healthcare strategies. The dataset is available at <https://www.kaggle.com/datasets/sulianova/cardiovascular-disease-dataset>.

3.3.5. Machine Learning-Generated Features

Machine learning-generated features are critical in improving the comprehensiveness of heart disease prediction models. Derived features are generated using advanced machine learning algorithms based on intricate patterns identified in the collected data. These derived features can encapsulate complex relationships, interactions, and nonlinearities in the dataset that traditional feature extraction methods may miss. Machine learning models' ability to recognize intricate patterns contributes to a deeper comprehension of the factors influencing heart disease. By incorporating these derived features into the predictive analysis, the model gains a better understanding of subtle indicators and relationships that contribute to accurate and personalized predictions. This method uses machine learning to unearth latent insights, ultimately improving the accuracy of heart disease prediction and supporting proactive healthcare interventions.

3.3.6. Time-domain and frequency-domain features

Important components in the comprehensive analysis of heart disease prediction models. A critical time-domain feature is RR interval variability, which represents variations in the intervals between consecutive heartbeats. The distance between R-R intervals within the signal is commonly used to calculate the heart rate of each signal. This R-R interval analysis is a fundamental method for assessing the frequency

and regularity of cardiac cycles, providing useful information for a variety of clinical applications such as heart rate variability analysis and arrhythmia detection [28,29]. RR interval variability analysis provides valuable insights into heart rate variability, a key indicator of cardiac health. The time-domain and frequency-domain characteristics of heart disease prediction models are critical components in the comprehensive analysis. RR interval variability, which represents variations in the intervals between consecutive heartbeats, is a critical time-domain feature. The addition of these time-domain and frequency-domain features improves the model's ability to detect subtle changes in heart dynamics, improving the accuracy and reliability of heart disease predictions. The combination of these features allows for a more comprehensive assessment of cardiac health, providing valuable information to healthcare professionals for personalized patient care.

$$SD_{RR} = \sqrt{\frac{\sum_{i=1}^N (RR_i - \overline{RR})^2}{N - 1}} \quad (6)$$

The power spectral density (PSD) is a key frequency-domain feature derived from the RR interval time series' Fourier transform. The distribution of energy in the signal is reflected in the power in different frequency bands. The power is commonly calculated in frequency bands associated with various physiological phenomena. These formulas provide quantitative measures of time-domain and frequency-domain features that capture important aspects of assisting in cardiac health assessment and disease prediction.

3.3.7. Statistical Features

Mean (μ)

$$\mu = \frac{\sum_{i=1}^N X_i}{N} \quad (7)$$

Standard deviation (σ)

$$\sigma = \sqrt{\frac{\sum_{i=1}^N (X_i - \mu)^2}{N}} \quad (8)$$

These statistical features provide critical insights into the data's central tendency, spread, and distribution, providing useful information for heart disease prediction and diagnosis.

3.4. TabNet

By adeptly handling structured datasets inherent in this domain, TabNet, a robust “Tabular Neural Network,” emerges as a powerful tool for predicting heart disease. Its application begins with meticulous data preparation, in which features relevant to heart health are organized in a tabular format and encoded to TabNet's specifications. The model's exceptional feature selection capabilities become critical, automatically determining the most important factors for predicting the presence or absence of heart disease. TabNet excels at learning intricate patterns indicative of cardiac health during the training phase on labeled datasets. Its distinct interpretability feature enables healthcare professionals to comprehend the key features influencing predictions, fostering transparency in critical decision-making. Following training, the model becomes adept at predicting the likelihood of heart disease for new data, which aids in risk assessment. Fine-tuning and validation improve the accuracy and reliability of the model, ensuring its applicability in real-world scenarios. Combining TabNet's strengths in handling tabular data and providing interpretable insights with advanced techniques such as CatBoost or IoMT has enormous potential for improving cardiovascular health predictions.

Fig. 3 depicts the architecture of the TabNet model, highlighting its unique design tailored for efficient tabular data processing. TabNet is built around encoder and decoder blocks, each of which plays an important role in feature learning and interpretation. The encoder uses a

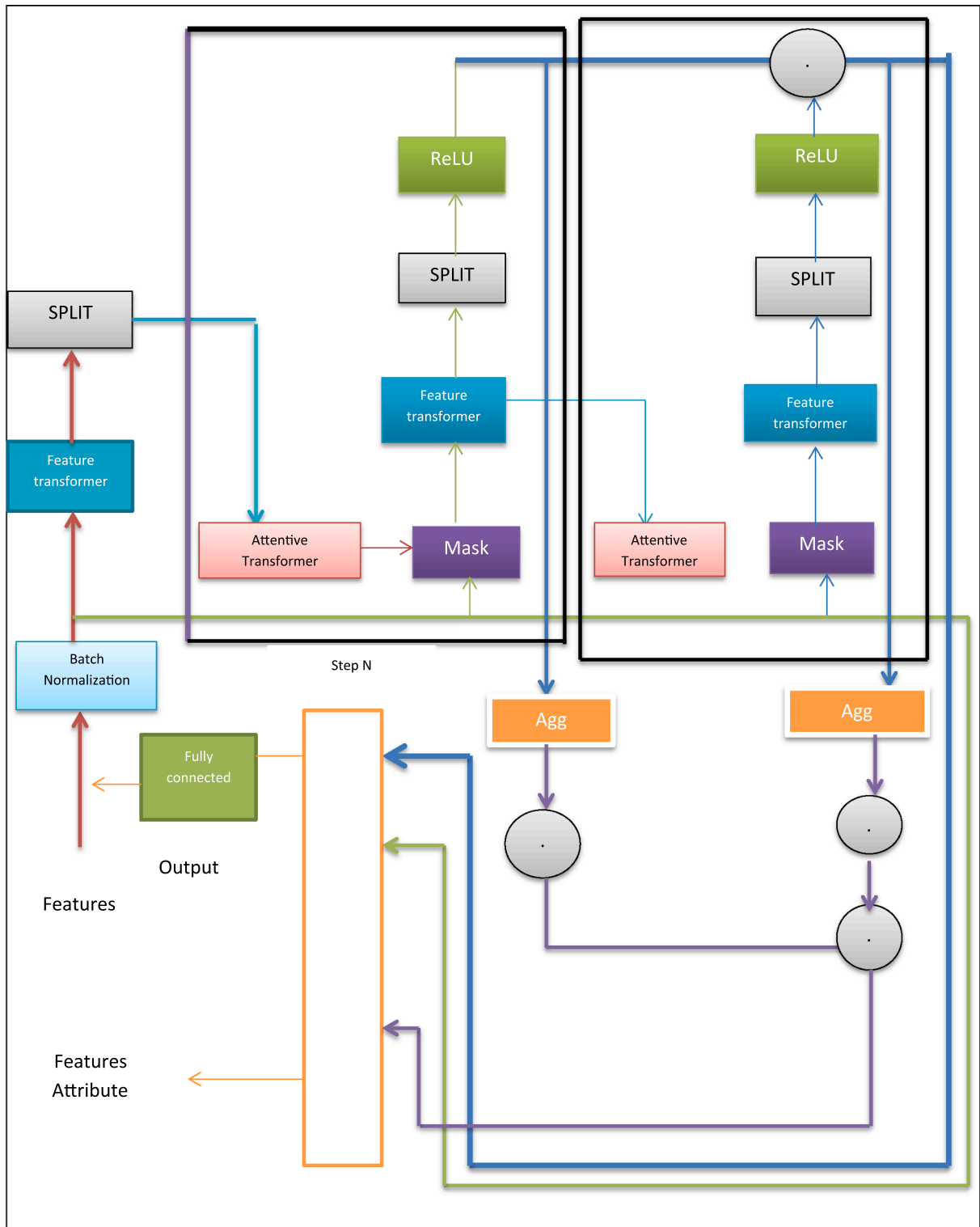


Fig. 3. TabNet Model Architecture.

sequential attention mechanism to detect informative patterns in the input features, while the decoder refines the learned representations for accurate predictions. TabNet can focus on relevant features thanks to this attention-based approach, which contributes to its interpretability and robust performance. The presence of shared and individual decision steps in the architecture improves its ability to detect important patterns.

$$GLUblock = FullyConnected - BatchNormalization - GLU(GatedLinearUnit) \quad (9)$$

$$GLU(X) = \sigma(x) \cdot x \quad (10)$$

$$\sum_{i=1}^n Sparsemax(x_i)_i = 1 \forall x \in R^n \quad (11)$$

An attention mask is an important component in neural network ar-

chitectures, especially those that use attention mechanisms such as the Transformer model. The attention mask aids in the control of information flow during self-attention or multi-head attention processes. The attention mask in the Transformer model is typically a square matrix with dimensions equal to the length of the input sequence. It contains values that indicate which sequence positions should be attended to and which should be ignored.

$$\eta_b[i] = \sum_{c=1}^{N_d} \text{ReLU}(d_{b,c}[i]) \quad (12)$$

$$M_{agg-b,j} = \frac{\sum_{i=1}^{N_{steps}} \eta_b[i] M_{b,j}[i]}{\sum_{j=1}^D \sum_{i=1}^{N_{steps}} \eta_b[i] M_{b,j}[i]^2} \quad (13)$$

3.4.1. TabNet Decoder

Fig. 4 depicts the TabNet Decoder Architecture in greater detail. The TabNet model's decoder architecture is a critical component responsible for generating predictions based on the information processed and learned in the encoder phase. The Decoder is made up of a series of steps that transform the encoded data into the final output. The feature transformer block, which consists of multiple GLU (Gated Linear Unit) blocks, is crucial. Each GLU block is made up of distinct layers that aid in the transformation process. The Feature Transformer is designed to extract relevant features for accurate predictions by capturing intricate patterns and dependencies within the input data.

$$\sum_{b=1}^B \sum_{j=1}^D \left| \frac{\hat{f}_{b,j} - f_{b,j} S_{b,j}}{\sum_{b=1}^B (f_{b,j} - 1/B \sum_{b=1}^B f_{b,j})^2} \right| \quad (14)$$

3.5. CatBoost

Because of its capacity to handle categorical characteristics and its robust gradient-boosting framework, CatBoost emerges as a key player in the field of heart disease prediction. CatBoost's categorical feature handling capabilities streamline the preprocessing pipeline, eliminating the need for extensive transformations and allowing the model to seamlessly integrate the various types of patient information found in heart disease datasets. Its mastery of the gradient boosting algorithm ensures that the model not only captures intricate patterns in the data but also continuously refines its predictions, resulting in a highly accurate and dependable predictive tool. CatBoost's resistance to overfitting is especially useful in healthcare applications, where generalizability to previously unseen patient profiles is critical. CatBoost mitigates overfitting concerns by utilizing depth-wise growth and a symmetric tree structure, ensuring that the heart disease prediction model retains its efficacy across diverse scenarios. Furthermore, the model's interpretability is an important asset in the healthcare landscape because it provides healthcare professionals with clear insights into the factors influencing predictions. Its ability to handle missing data, support cross-validation, and allow for hyperparameters tuning adds to its versatility, allowing for a tailored and optimized approach to heart disease prediction. CatBoost's integration into a holistic predictive healthcare model holds the promise of a comprehensive and accurate tool for cardiovascular health monitoring and risk assessment, as it naturally aligns with ensemble learning principles.

$$B(x) = \sum_{k=1}^K \alpha_k \cdot T_k(x) \quad (15)$$

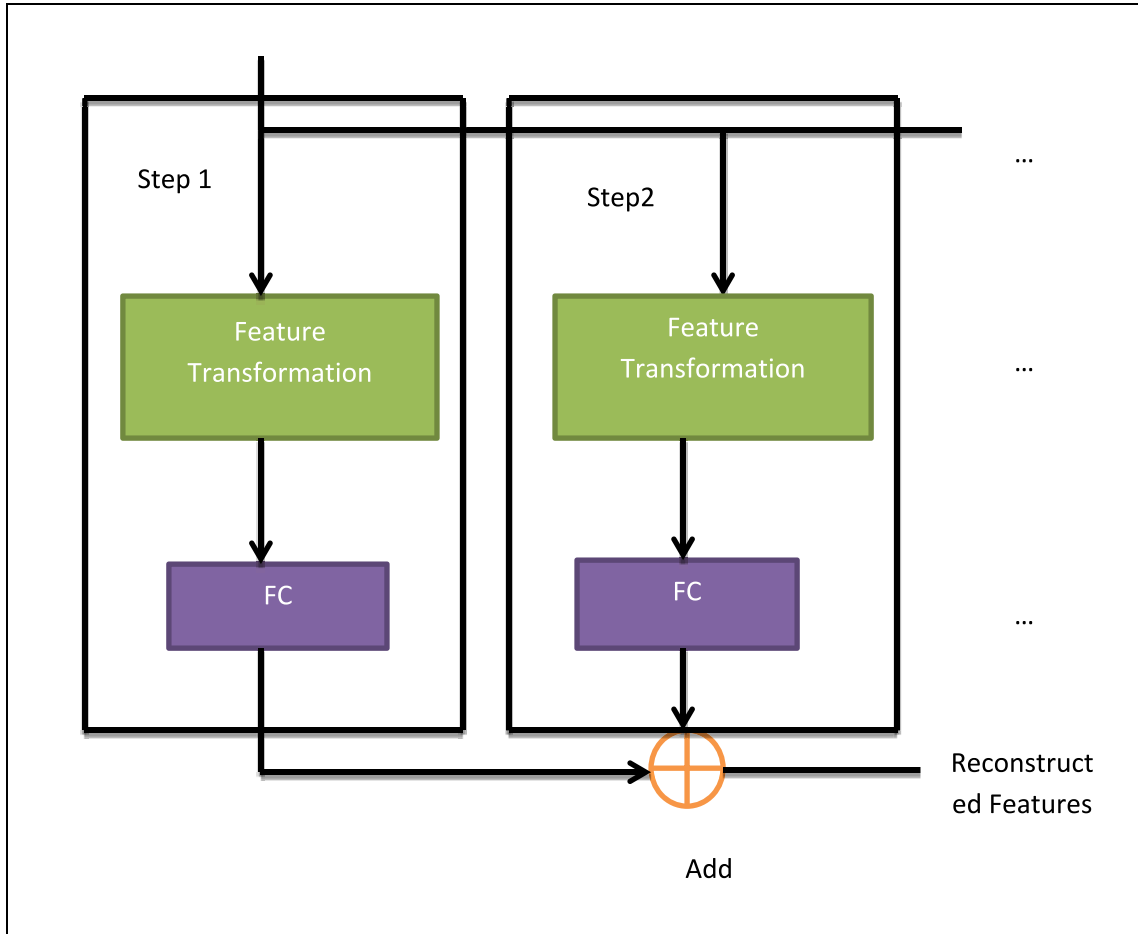


Fig. 4. TabNet Model Decoder Architecture.

Fig. 5 depicts CatBoost's architecture, as well as a visual representation of its underlying mechanisms for heart disease prediction. CatBoost is based on a gradient boosting framework, in which an ensemble of decision trees collaborates to refine predictions. The architecture demonstrates the hierarchical structure of decision trees, with each acting as a weak learner that contributes to the model's overall predictive strength. CatBoost's distinguishing feature is its ability to handle categorical features adeptly, as illustrated in the architecture by the seamless integration of categorical variable nodes within the trees. This addresses a common issue in heart disease prediction datasets where variables like gender or type of chest pain are categorical. The architecture of CatBoost demonstrates the efficient conversion of categorical features into numerical representations. Furthermore, the architecture visually emphasizes CatBoost's ensemble learning nature, in which multiple decision trees collaborate to improve predictive accuracy. The architecture emphasizes the model's focus on instances where misclassifications occurred in previous iterations, which contributes to continuous model improvement.

3.6. Software implementation

Utilizing TabNet and CatBoost for heart disease prediction involves leveraging Python-compatible software and libraries for machine learning model development and training. The computational execution of the models occurred on a single Windows10-based computer (HP ProBook 450G) equipped with an Intel (R) Core (TM) i5-4200 M CPU @ 2.50 GHz and 8 GB RAM. Python's ecosystem includes powerful data manipulation, model implementation, and evaluation tools. Popular libraries such as pandas, NumPy, scikit-learn, PyTorch, and TabNet-specific libraries (e.g., pytorch_tabnet) and CatBoost make the process easier. To begin, make sure you have a Python environment set up and

install any required libraries with the package manager pip. Once installed, load your heart disease dataset into a Pandas DataFrame. Import the necessary libraries for TabNet, load and split the dataset, initialize the TabNet Classifier, and train the model. Evaluate its performance on the test set and investigate feature significance. To achieve the best results, fine-tune the hyperparameters. Similarly, for CatBoost, use data preparation libraries such as pandas, NumPy, and scikit-learn. Import the CatBoostClassifier, then initialize, train, and evaluate its performance. CatBoost is a valuable addition to the predictive modeling pipeline because of its inherent ability to handle categorical features, manage overfitting, and provide interpretability. Integrating these models into a healthcare environment entails taking into account real-time prediction capabilities for streaming data as well as ensuring compatibility with an Internet of Medical Things (IoMT) framework. Once the models' performance has been validated, deploy them in healthcare settings while adhering to security and privacy standards. TabNet and CatBoost synergy, combined with appropriate software implementation, contribute to a robust and accurate heart disease prediction system.

Pseudo code for feature extraction and prediction

Algorithm

```

Step1: Import necessary libraries and modules
Step2: Load the clinical dataset
pd.read_csv("clinical_dataset.csv")
Step3: Data preprocessing, value reduction handling, and categorized feature encoding
Step 4: Separate the data into features (X) and target variables (y).
Step 5: Divide the data into training and testing groups.
Step 6: Get the TabNet model ready for feature selection.
tabnet_model = TabNetClassifier()
Step7: Train the TabNet model on the training data

```

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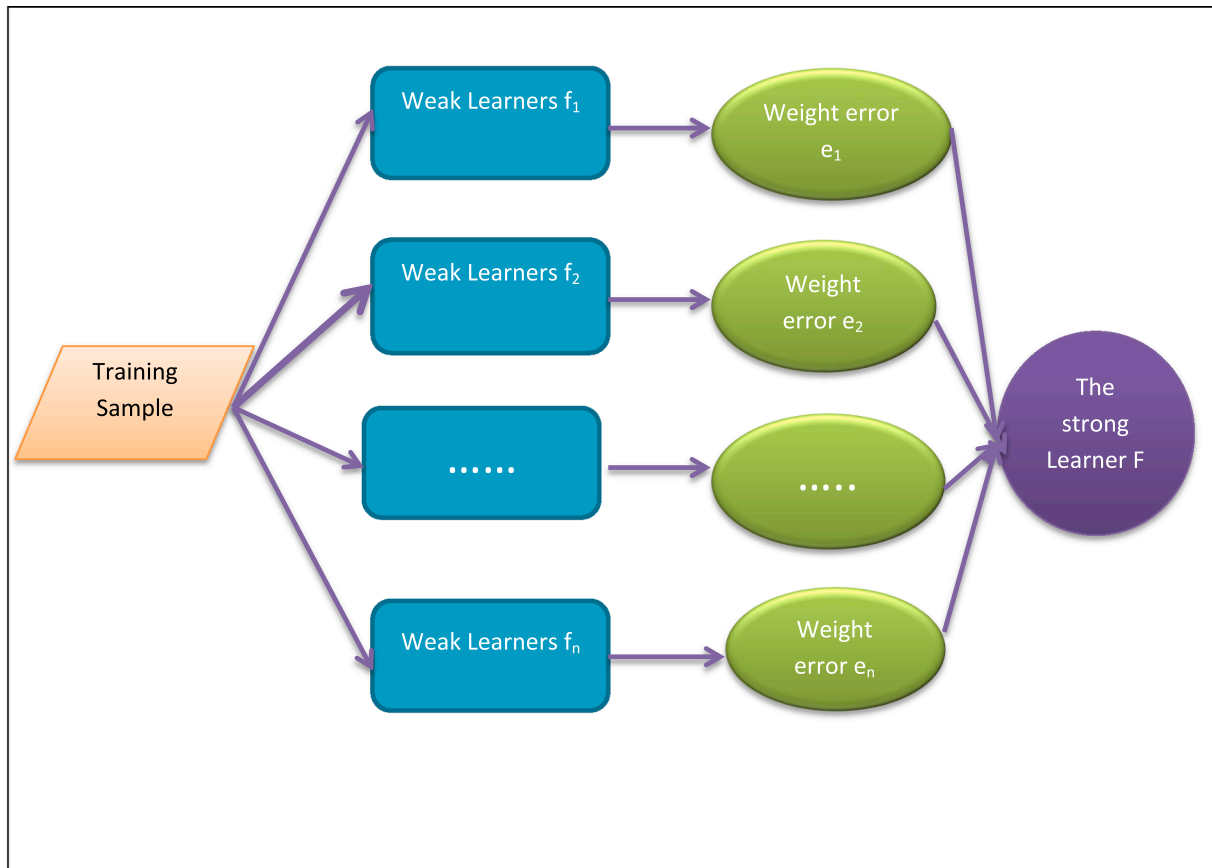


Fig. 5. CatBoost Architecture.

(continued)

Algorithm
Step8:Select important features using TabNet
Step 9:Extract the selected features from the dataset
Step10:Initialize the catBoost model for predictive accuracy
Step 11: Train the catBoost model on the selected features
Step 12:Make predictions on the test set
Step 13:Evaluate the model performance
Step14: Output the results

4. Result and discussion

The integration of IoMT and AI in heart disease prediction model is a game changer in cardiovascular health. The model's ability to seamlessly integrate data from a variety of medical devices via IoMT, combined with the precision of the TabNet and catBoost algorithms, has resulted in a predictive tool that goes above and beyond traditional approaches. Individuals and healthcare providers can use personalized risk assessments and real-time predictions to adopt proactive strategies, moving away from reactive healthcare models. Looking ahead, this convergence has the potential to redefine preventive care and reduce the global burden of cardiovascular diseases. Because the model emphasizes the importance of systemic efforts in healthcare policy and education, it contributes to a paradigm shift in how societies approach cardiovascular wellness. This integrated approach, with its promise of improved patient outcomes and informed preventive measures, represents a significant step toward a future in which technology and healthcare intersect to create a more resilient and proactive system.

4.1 Performance metrics

These key performance metrics provide a solid framework for assessing the efficacy and impact of innovative cardiovascular wellness model. The emphasis on accuracy, precision, recall, F1 score, ROC-AUC, and MSE reflects a thorough evaluation strategy that ensures the model's ability to deliver reliable predictions and contribute to proactive cardiovascular health management. Precision and recall metrics address the delicate balance of reducing false positives while capturing true positives, both of which are critical considerations in identifying individuals at risk of heart disease. As a harmonic mean, the F1 score encapsulates the model's overall effectiveness, especially when dealing with imbalanced class distributions. Furthermore, taking execution time into account recognizes the practical implications of real-time

predictions, particularly in clinical settings where prompt interventions are critical. The model's accuracy and relevance in real-world applications will require continuous monitoring and refinement of these performance metrics. By adhering to these standards, the cardiovascular wellness model positions itself as a trailblazer at the intersection of IoMT and AI, with the potential to significantly advance proactive healthcare strategies for heart disease prevention and management.

4.2. Comparative existing models

The presented model is distinguished by its seamless integration of the IoMT with advanced AI techniques, specifically the TabNet and catBoost algorithms. This comprehensive approach improves the model's ability to collect data from various medical devices and process complex physiological data for accurate predictions.

Fig. 6 compares the accuracy of the existing model to that of the TabNet with CatBoost-enhanced model. This visual representation provides critical insight into the performance improvements realized by incorporating advanced machine learning techniques. Starting with accuracy, the proposed TabNet with CatBoost model achieves an exceptional accuracy of 98 %, showcasing a substantial advancement over the existing models—SVM (74.5 %), HSC (85.2 %), and RCNN (95.5 %). This remarkable accuracy underscores the effectiveness of the novel approach in accurately predicting heart disease. The accuracy of the existing model is compared to the outcomes of the enhanced model, demonstrating the efficacy of TabNet and CatBoost in improving predictive accuracy.

$$\text{Accuracy} = (TP + TN) / (TP + TN + FP + FN)$$

(16)

Fig. 7 depicts a valuable glimpse into the nuanced aspects of model performance, particularly its ability to detect and capture positive cases. In terms of sensitivity, which measures the ability to correctly identify positive cases, the proposed system excels with a sensitivity of 97.1 %. This outperforms the existing models—SVM (74.8 %), HSC (89.0 %), and RCNN (96.7 %). The higher sensitivity implies a superior capacity to detect individuals with heart disease, reducing the likelihood of false negatives. This comparison is essential for understanding the nuanced impact of using advanced algorithms like TabNet with CatBoost on the predictive model's sensitivity.

$$\text{Recall} = TP / (TP + FN)$$

(17)

Fig. 8 depicts the models' nuanced abilities in accurately identifying

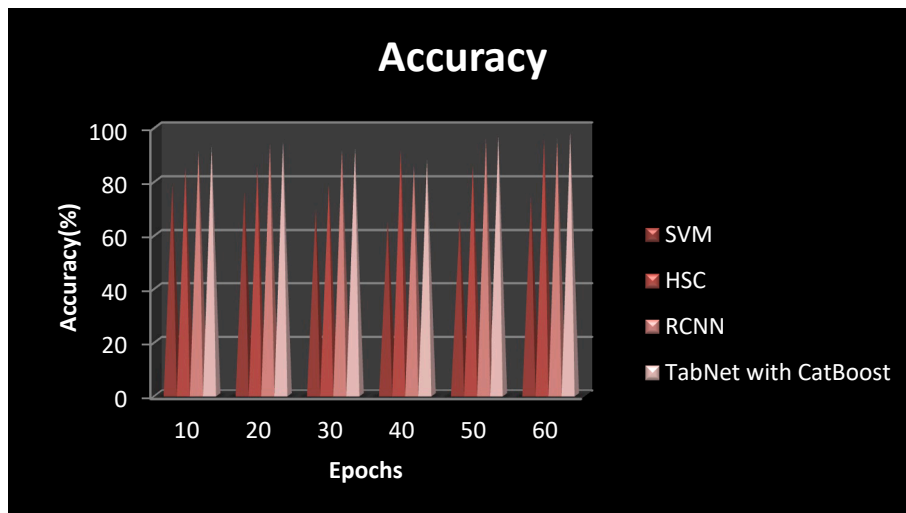


Fig. 6. Comparison between the Existing and TabNet with CatBoost Accuracy.

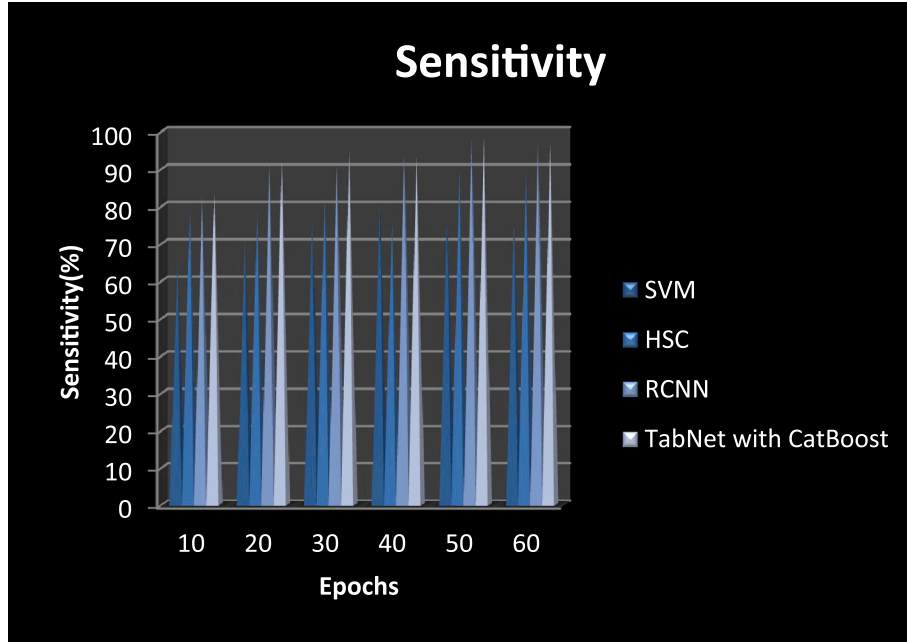


Fig. 7. Comparison between the Existing and TabNet with CatBoost Sensitivity.

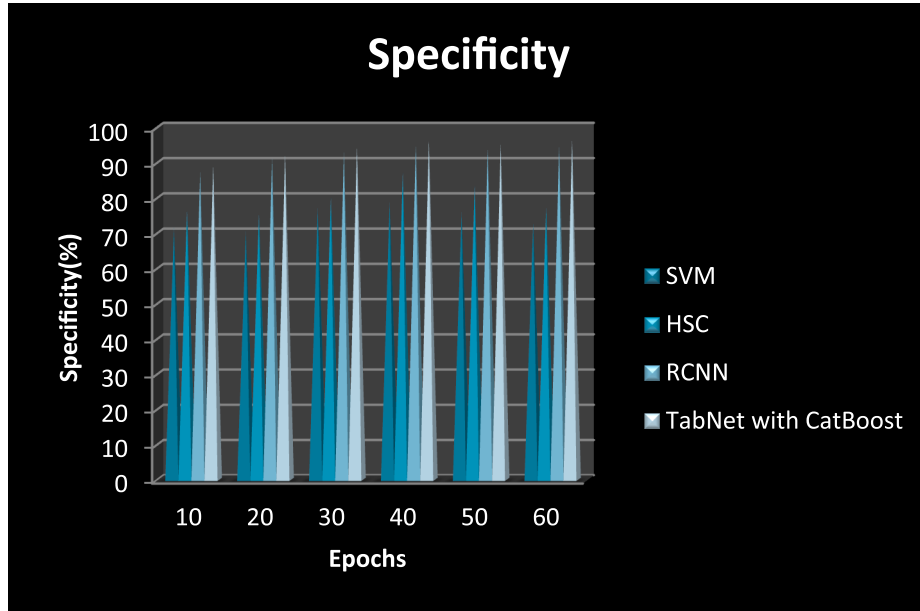


Fig. 8. Comparison between the Existing and TabNet with CatBoost Specificity.

negative cases. Specificity, representing the ability to correctly identify negative cases, is another area where the proposed system demonstrates superiority. With a specificity of 96.5 %, it surpasses SVM (73.1 %), HSC (77.44 %), and RCNN (94.6 %). The enhanced specificity indicates a reduced rate of false positives, contributing to the reliability of the proposed system.

$$Specificity = \frac{TN}{TN + FP} \quad (18)$$

Fig. 9 shows a detailed comparison of the precision of the existing model and the TabNet with CatBoost-enhanced model. Precision, a metric assessing the accuracy of positive predictions, is notably higher in the proposed system at 95.3 %, compared to SVM (73.0 %), HSC (83.1 %), and RCNN (92.3 %). This signifies a reduction in the occurrence of false

positive predictions, highlighting the precision of the novel approach.

$$Precision = TP / (TP + FP) \quad (19)$$

Fig. 10 depicts a comprehensive comparison of the error rates of the existing model and the TabNet with CatBoost-enhanced model. The error rate is a critical metric that captures the overall accuracy of predictive systems, taking into account both false positives and false negatives. The error rate is significantly lower in the proposed system at 25.6 %, marking a substantial improvement over SVM (57.2 %), HSC (39.3 %), and RCNN (31.0 %). A lower error rate indicates a higher overall accuracy in predictions, reinforcing the reliability of the proposed model.

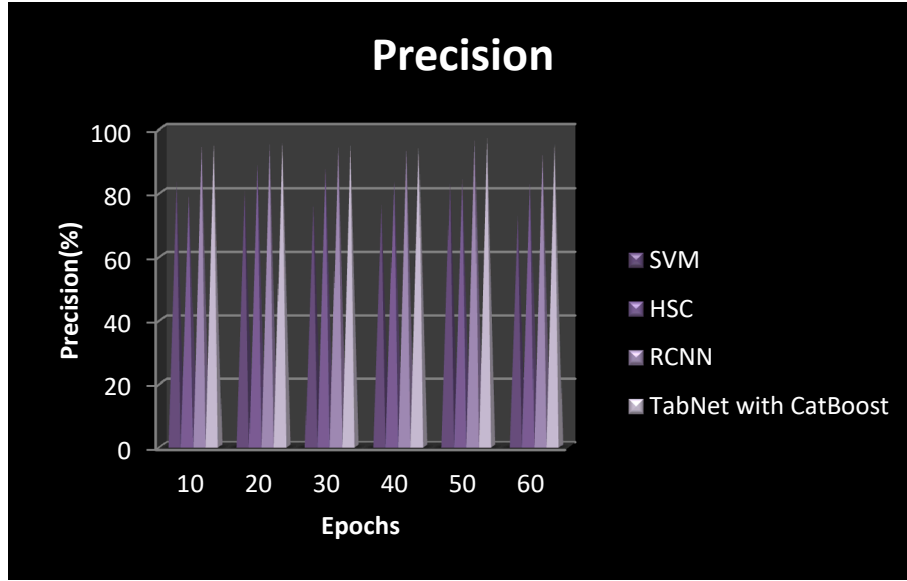


Fig. 9. Comparison between the Existing and TabNet with CatBoost Precision.

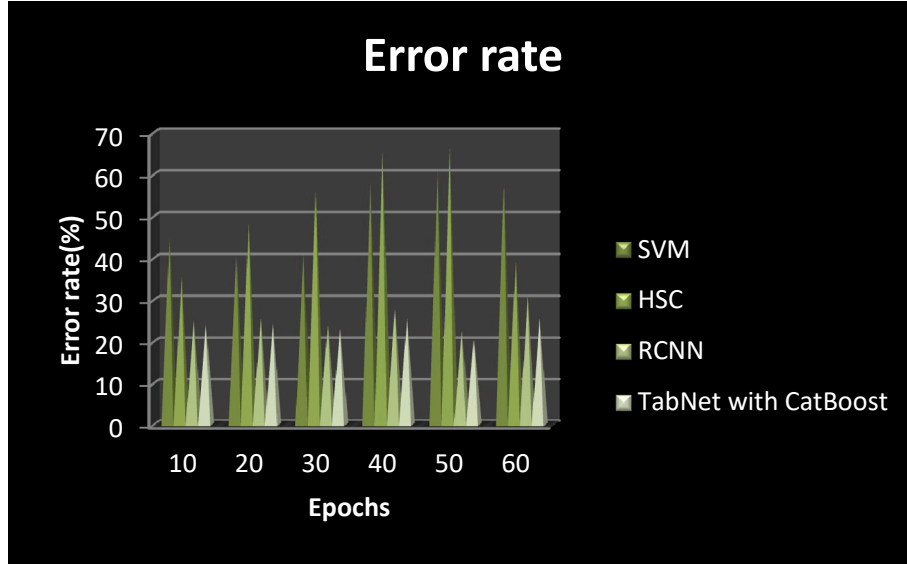


Fig. 10. Comparison between the Existing and TabNet with CatBoost Error Rate.

$$ErrorRate = \frac{1}{n} \sum_{i=0}^n \frac{TrueNegative_i + FalseNegative_i}{TotalNo.ofSamples} \quad (20)$$

Fig. 11 shows a detailed comparison of the processing times of the existing model and the TabNet with CatBoost-enhanced model. Processing time is an important metric that reflects the models' computational efficiency in making predictions. One of the critical factors influencing the practical applicability of these models is processing time. The proposed TabNet with CatBoost model achieves a processing time of 26.4 units, demonstrating enhanced efficiency compared to SVM (61.0), HSC (47.7), and RCNN (32.7). The reduction in processing time is crucial for real-time applications and contributes to the practical feasibility of the proposed system.

Table 1 compares the existing system, which includes Support Vector Machine (SVM), Hierarchical Self-Calibrating (HSC), and the Region Convolutional Neural Network (RCNN), to the proposed TabNet with CatBoost algorithm in the context of cardiovascular wellness.

Fig. 12 evaluates evaluation metrics for existing models and the

proposed TabNet with the CatBoost algorithm. The metrics shown in the figure give a comprehensive picture of each model's ability to predict heart disease. In summary, the proposed TabNet with CatBoost model outperforms existing models across all assessed metrics, showcasing its efficacy in heart disease prediction. The combination of TabNet and CatBoost yields a highly accurate, sensitive, and specific predictive model with improved precision and reduced error rates. Additionally, the proposed system exhibits efficiency gains, as evidenced by its shorter processing time. These advancements collectively position the proposed system as a robust and efficient solution for heart disease prediction, suggesting its potential for practical implementation in healthcare settings. The superiority of the proposed system can be attributed to the synergistic benefits of TabNet and CatBoost. TabNet's ability to handle structured datasets, interpretability, and feature selection contributes to its effectiveness in capturing intricate patterns relevant to heart health. CatBoost's strength in handling categorical features, mitigating overfitting, and providing interpretability aligns well with the challenges posed by heart disease datasets. The ensemble nature of CatBoost,

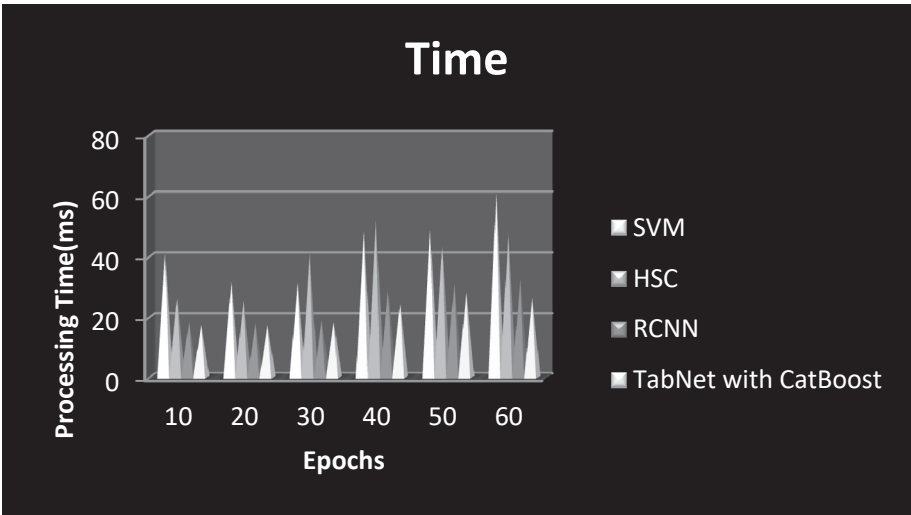


Fig. 11. Comparison between the Existing and TabNet with CatBoost Processing time.

Table 1
Comparison between the Existing and TabNet with CatBoost.

Existing System			Proposed System	
Models	SVM	HSC	RCNN	TabNet with CatBoost
Accuracy	74.5	85.2	95.5	98
Sensitivity	74.8	89.0	96.7	97.1
Specificity	73.1	77.44	94.6	96.5
Precision	73.0	83.1	92.3	95.3
Error rate	57.2	39.3	31.0	25.6
Time	61.0	47.7	32.7	26.4

combining multiple decision trees, enhances its predictive accuracy. While SVM, HSC, and RCNN have demonstrated utility in various contexts, their performance falls short when compared to the integrated approach of TabNet with CatBoost. The novel system leverages the strengths of each component to address the complexity and nuances of heart disease prediction. The comprehensive comparison presented underscores the potential of the proposed system to revolutionize

predictive modeling in the domain of cardiovascular health. In conclusion, the proposed system emerges as a transformative solution, offering advancements in accuracy, sensitivity, specificity, precision, error rate, and processing time over existing models. This research lays the foundation for a state-of-the-art predictive healthcare model that can significantly enhance the early detection and management of heart diseases, ultimately contributing to improved patient outcomes.

Discussion

The proposed system introduces a unique and comprehensive approach to predicting heart disease by seamlessly integrating IoMT with advanced AI techniques, specifically TabNet and CatBoost algorithms. This novel combination enhances the model's capability to collect diverse physiological data from medical devices, improving accuracy in predicting heart disease. TabNet's adept handling of structured datasets, interpretability, and feature selection, combined with CatBoost's prowess in managing categorical features, mitigating overfitting, and providing interpretability, sets this system apart. Unlike existing models (SVM, HSC, RCNN), our approach achieves exceptional accuracy (98 %), sensitivity (97.1 %), and specificity (96.5 %),

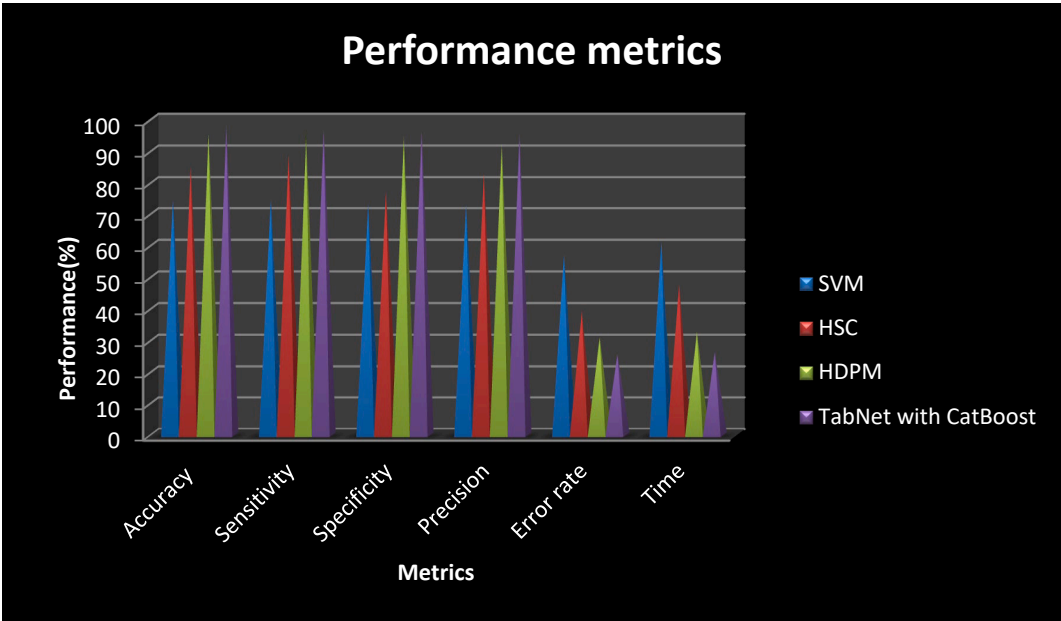


Fig. 12. Evaluation metrics for Existing and TabNet with CatBoost.

showcasing superior performance across critical metrics. The precision of 95.3 % and a lower error rate of 25.6 % further underscore its effectiveness. Despite these strengths, challenges include the need for large and diverse datasets for training, potential biases in data, and interpretability of complex machine learning models in a healthcare context. Ensuring real-time processing efficiency and addressing data privacy concerns also pose implementation challenges. The system's performance is contingent on continuous updates to accommodate evolving medical knowledge. While limitations exist, such as the requirement for robust cybersecurity measures, our proposed system represents a transformative advancement in predictive healthcare modeling for heart disease, with potential implications for personalized patient care and improved outcomes.

5. Conclusion and future scope

The study introduces a groundbreaking Coronary Artery Disease Forecast Model, leveraging TabNet and CatBoost algorithms, marking a paradigm shift in heart disease prediction. This innovative model not only enhances accuracy and efficiency but also facilitates real-time data acquisition through IoMT, promising transformative impact. TabNet's role in feature selection and extraction, coupled with CatBoost's predictive accuracy, improves precision and interpretability. Envisioned as a transformative tool in healthcare, the model integrates IoMT and advanced computer vision, revolutionizing predictive healthcare. Despite exceptional accuracy (98 %) in our numerical results, limitations include dataset constraints and algorithm-specific considerations. Future work involves adapting the model to larger, diverse datasets, aligning with evolving healthcare landscapes, and integrating emerging technologies like 5G and blockchain. Clinical validation, collaborative implementation with healthcare institutions, and ensuring an accessible, ethical interface are crucial for responsible adoption. Exploring additional predictive metrics and extending capabilities for long-term monitoring positions the model beyond heart disease, contributing to comprehensive patient care. The convergence of IoMT, wearable technology, and AI underscores the model's importance. As it evolves, the model is poised to revolutionize proactive global cardiovascular health management, emphasizing the critical need for continuous refinement and ethical considerations in its widespread adoption.

CRedit authorship contribution statement

K.K. Baseer: Resources. **K. Sivakumar:** Resources. **Duggineni Veeraiah:** Resources. **Gunjan Chhabra:** Resources. **Prasanna Kumar Lakineni:** Resources. **M. Jahir Pasha:** Resources. **Ramu Gandikota:** Resources. **Gopakumar Hari Krishnan:** Resources.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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